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An Integrated Full Consistency Method–Data Envelopment Analysis Framework for Evaluating Logistics Process Efficiency in the Process Industry: Evidence From an Oil Refinery

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Abstract: Efficient logistics processes are essential to the operational performance of process-industry companies, where material receipt, warehousing, and internal distribution are closely connected with production continuity. However, evaluating these processes remains difficult because logistics performance depends on multiple criteria with different levels of importance, while conventional assessments often provide limited information about the relative efficiency of individual logistics units. This study investigates the efficiency of logistics activities through an integrated decision-analytics framework combining the Full Consistency Method (FUCOM) and Data Envelopment Analysis (DEA). FUCOM was used to determine the relative importance of the evaluation criteria, and DEA was applied to measure the technical efficiency of the logistics units. The framework was examined through a case study of the Modriča Oil Refinery, covering material receipt, warehousing, and the internal distribution of products to production facilities. The results showed clear differences in efficiency among the analyzed logistics segments and identified the processes that represented critical points within the logistics system. The sensitivity analysis further confirmed the stability of the evaluation results under changes in the criteria weights. These findings indicate that combining criterion weighting with efficiency measurement provides a structured basis for identifying operational weaknesses and setting improvement priorities. The proposed framework supports data-informed logistics decisions in the process industry and can be adapted to comparable industrial settings involving interconnected logistics activities.

Keywords: Logistics process efficiency; Decision analytics; Full Consistency Method; Data Envelopment Analysis; Process industry; Logistics performance; Operational decision support

1 Introduction

The efficiency of logistics processes plays a central role in the competitiveness of industrial companies. Procurement, warehousing, material handling, and distribution account for a considerable share of operating costs, while companies face growing demands for reliable supply, timely delivery, and sustainable operations. In modern industrial systems, logistics is no longer limited to supporting production. It is also a strategic function that directly affects profitability, production continuity, service quality, and supply chain resilience. Its importance is particularly evident in the process industry, where the flows of raw materials, intermediate products, and finished products are closely interconnected. A delay or disruption at one stage may therefore affect the entire production system.

Logistics processes in raw-material processing industries are characterized by complex material flows, changing demand, limited storage capacity, and exposure to external disturbances. Decision-making in this setting is further complicated by the nature of the available information, which may include imprecise data, approximate estimates, and subjective expert judgments. Conventional efficiency assessments are commonly based on separate performance

indicators or financial measures. Although such measures provide useful information about individual activities, they do not offer a sufficiently comprehensive account of the operational, organizational, and managerial dimensions of logistics performance. They also provide a limited basis for comparing logistics units that use different combinations of resources to produce multiple outputs.

Multi-criteria decision-making (MCDM) methods provide an appropriate analytical basis for addressing this problem because they allow several interconnected and potentially conflicting criteria to be considered simultaneously. However, the reliability of an MCDM assessment depends on how expert preferences are elicited and how consistently the relative importance of the criteria is determined. Existing studies do not always give sufficient attention to the imprecision and inconsistency associated with expert judgments. Moreover, criterion weighting alone does not establish whether the observed logistics units use their available resources efficiently. A complete evaluation therefore requires a methodological framework that combines structured expert assessment with an objective measurement of operational performance.

This need is especially relevant to the oil refining industry. Refinery operations depend on the continuous receipt of raw materials, the appropriate use of storage capacity, and the timely internal distribution of materials and products to production facilities. Despite the operational importance of these activities, there remains a lack of integrated models specifically developed to evaluate the efficiency of internal logistics processes in oil refining. Previous Full Consistency Method (FUCOM)–Data Envelopment Analysis (DEA) applications have examined suppliers, transport networks, and logistics service providers, but the use of this combined approach for internal logistics activities in the process industry remains limited. In particular, few studies have integrated subjective expert judgments with objective operational data in a way that supports both efficiency measurement and the identification of improvement priorities under variable industrial operating conditions.

This study investigates logistics process efficiency by developing and applying an integrated FUCOM–DEA decision-analytics framework. The FUCOM was used to determine the relative importance of the evaluation criteria from expert assessments while maintaining consistency in the weighting procedure. DEA was then used to measure the technical efficiency of the observed logistics units on the basis of defined input and output variables. Rather than using the FUCOM results only to rank the criteria, the resulting weights were incorporated into the DEA model through Assurance Region (AR) restrictions. This integration constrained the flexibility of the DEA weights according to expert-defined operational priorities and produced an efficiency assessment that more closely reflected the decision environment of the logistics system.

The proposed framework was examined through a case study of the Modriča Oil Refinery, which represents the process industry through its complex logistics flows and its dependence on an uninterrupted supply to production. The empirical analysis covered three central areas of internal logistics: raw material receipt, warehousing, and the internal distribution of products to production facilities. The analysis was designed to compare the technical efficiency of the observed logistics units, identify critical points in the logistics system, and determine the principal factors associated with logistics performance. A sensitivity analysis was also conducted to examine the stability of the model and the robustness of the efficiency results.

The contribution of this study is threefold. First, it develops an integrated framework that connects consistent expert-based criterion weighting with the objective measurement of logistics process efficiency. Second, it embeds FUCOM-derived weight information within the DEA model through AR restrictions, rather than treating criterion weighting and efficiency measurement as separate analytical stages. Third, it applies this framework to internal logistics operations in the oil refining industry, an area that has received limited attention in previous FUCOM–DEA research. The resulting analysis provides a structured basis for locating operational weaknesses, setting improvement priorities, and supporting logistics management decisions. Because the framework relies on criteria, inputs, and outputs that can be adjusted to a specific operating environment, it may also be applied to other process-industry systems with comparable logistics structures.

2 Literature Review

The assessment of logistics process efficiency is an established research area in logistics and supply chain management because logistics performance directly affects operating costs, resource utilization, and corporate competitiveness. Efficiency measurement is particularly important in the process industry, where logistics operations include raw material procurement, warehousing, internal material handling, and the distribution of finished products. These activities involve interconnected material flows and multiple inputs and outputs, making their performance difficult to represent through a single financial or operational indicator. Quantitative methods capable of comparing several dimensions of performance are therefore required.

DEA is among the most widely used methods for benchmarking organizational and logistics performance. According to Cooper et al. [1], DEA measures the relative efficiency of homogeneous decision-making units (DMUs) by considering multiple inputs and outputs without requiring a predefined functional relationship between them. This feature makes DEA suitable for logistics systems in which units consume different combinations of

resources and generate several operational outcomes. The method has consequently been applied to transportation, warehousing, supplier selection, logistics services, and supply chain management.

A considerable body of research has used DEA to evaluate logistics and transportation performance. Toloo and Nalchigar [2] employed a DEA model for supplier selection, whereas Adalı and Işık [3] developed an integrated Decision Making Trial and Evaluation Laboratory (DEMATEL)–Analytic Network Process (ANP)–DEA model for selecting third-party logistics providers. The development of hybrid DEA models can also be observed in railway transportation. Blagojević et al. [4] proposed an integrated Fuzzy Analytical Hierarchy Process (FAHP)–DEA approach for evaluating railway freight operators. Fuzzy AHP was used to select the relevant criteria, and DEA was then applied to measure the relative efficiency of the operators. Stević et al. [5] developed an integrated Principal Component Analysis (PCA)–DEA–MCDM model for evaluating transportation companies. Their framework combined PCA and DEA with the CRiteria Importance Through Intercriteria Correlation (CRITIC), Entropy, and Measurement Alternatives and Ranking according to Compromise Solution (MARCOS) methods. This multiphase procedure was used to identify efficient and inefficient operating periods. Taken together, these studies show how criteria selection and weighting procedures can be incorporated into DEA-based assessments of complex transportation systems.

The methodological development of DEA has also included super-efficiency formulations, such as those discussed by Tone [6], which allow efficient DMUs to be differentiated beyond the conventional efficiency frontier. In a separate line of research, Chai et al. [7] examined fuzzy MCDM methods in supplier selection and demonstrated how uncertainty and subjective assessments could be represented supply chain decisions. These developments provided part of the methodological basis for subsequent models combining MCDM procedures with DEA.

Despite its ability to measure relative efficiency, conventional DEA has a well-recognized limitation: each DMU is generally allowed substantial freedom in selecting the weights assigned to its inputs and outputs. A model may consequently assign very small or zero weights to indicators that decision makers consider operationally important. Although this flexibility allows each DMU to be evaluated under favorable conditions, it can also produce results that are difficult to reconcile with managerial priorities. This limitation has led to the development of integrated models in which MCDM methods provide information about criterion importance and DEA performs the efficiency measurement.

The FUCOM [8], is a more recent method for determining criterion weights. It requires relatively few pairwise comparisons and controls deviations from full consistency in the expert judgments. These characteristics make the method suitable when the number of available experts or the time available for eliciting judgments is limited. FUCOM has been applied in logistics, transportation, risk management, and the performance assessment of different organizational and technical systems.

Several studies have used FUCOM in logistics and performance management. Aka [9] has applied FUCOM method in fuzzy form to select material handling equipment in automotive sector. Abdullah et al. [10] and Khosravi et al. [11] applied FUCOM-MARCOS frameworks to the assessment of health-system performance. A previous research [12] aimed to identify the expectations of shippers and stakeholders regarding the use of unmanned aerial vehicles (UAVs) in logistics operations using FUCOM method.

A more directly relevant stream of research combines FUCOM with DEA. Starčević et al. [13] applied AHP-DEA approach for selection of a terrain vehicle, while FUCOM method has been applied in reliability models to validate previously obtained results. Mitrović Simić et al. [14] proposed a CRITIC–Fuzzy FUCOM–DEA–Fuzzy MARCOS model for road-section safety assessment, while Yürüyen et al. [15] applied Fuzzy FUCOM- to evaluate criteria by adding environmental parameters into the criteria for selection of transportation vehicles. Pajić et al. [16] developed a DEA–FUCOM–Combined Compromise Solution (CoCoSo) framework for procurement optimization and supplier selection. Although these studies share the use of FUCOM and DEA, they differ in how the weighting, efficiency measurement, and final ranking stages are connected.

Other hybrid frameworks have combined DEA with alternative MCDM procedures in logistics and supply chain applications. Le and Pham [17] developed a three-phase DEA–FAHP–Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (FTOPSIS) model for evaluating logistics companies. The model combined efficiency analysis with further procedures for determining criterion importance and ranking logistics companies. Wang et al. [18] proposed a Composite Indicators (CI)–DEA–Integrated Determination of Objective CRIteria Weights (IDOCRIW)–Multi-Attributive Border Approximation Area Comparison (MABAC) model for cross-dock terminal location selection, illustrating how DEA and MCDM methods can be connected in a complex logistics location problem. Petrović et al. [19] applied a three-phase DEA–Fuzzy AHP model to evaluate and rank logistics companies, while Yang et al. [20] used DEA to assess the relative efficiency of third-party logistics companies operating in road transportation and warehousing. These studies demonstrate that hybrid frameworks may serve different purposes, including variable reduction, efficiency measurement, criterion weighting, and the subsequent ranking of DMUs.

The reviewed literature indicates that integrating DEA with MCDM methods has become an important direction in the assessment of logistics and transportation systems. Such integration is especially relevant when managerial

information about the relative importance of the indicators needs to be considered alongside operational input and output data. MCDM methods can provide structured information about decision-maker preferences, while DEA can determine how efficiently individual units transform inputs into outputs. However, direct FUCOM–DEA integration remains less common than models combining DEA with more established weighting methods.

Existing FUCOM–DEA studies have mainly addressed transportation safety, supplier selection, procurement processes, and the efficiency of warehouse equipment. Their analytical focus has generally been placed on external suppliers, transport operators, infrastructure sections, or individual items of logistics equipment. The use of an integrated FUCOM–DEA model to evaluate internal logistics processes in the process industry has received much less attention. In particular, there is limited research treating raw material receipt, warehousing, and internal material distribution as interconnected logistics activities within a single process-industry system.

Based on this gap, the present study applies an integrated FUCOM–DEA model to evaluate logistics process efficiency in the process industry. FUCOM was used to determine the relative importance of the relevant criteria from expert assessments. The resulting priority information was then incorporated into the DEA model through AR restrictions, thereby limiting unrestricted weight selection and aligning the efficiency assessment with the operational priorities of the decision makers. This structure combines expert knowledge with quantitative efficiency measurement and provides a basis for identifying inefficient logistics activities and locating areas in which operational performance may be improved.

3 Methods

The assessment of the efficiency of logistics activities in the process industry is a complex analytical problem that includes a large number of interrelated criteria and operational parameters. In addition, the evaluation of logistics systems is often based on expert assessments that include a certain degree of uncertainty and subjectivity. Therefore, for the purposes of this research, an integrated methodological framework was developed that combines the FUCOM and DEA methods. After determining the weights of the criteria, the DEA method is applied to assess the relative efficiency of the observed logistics units based on the relationship between input and output variables. The integrated FUCOM–DEA approach allows for the combination of subjective expert assessments and quantitative operational data, thus providing a stable and methodologically consistent analysis of logistics performance.

3.1 Description of Research Stages

The methodology of this research is based on an integrated FUCOM–DEA approach that allows for the combination of expert assessments and quantitative analysis of logistics performance. The methodological procedure begins with defining logistics efficiency criteria, after which the FUCOM method is applied to determine their relative weights. In the next phase, DMUs are formed and input and output variables are defined, after which the DEA model is applied to assess relative efficiency. The methodological framework of the research is shown in Figure 1.

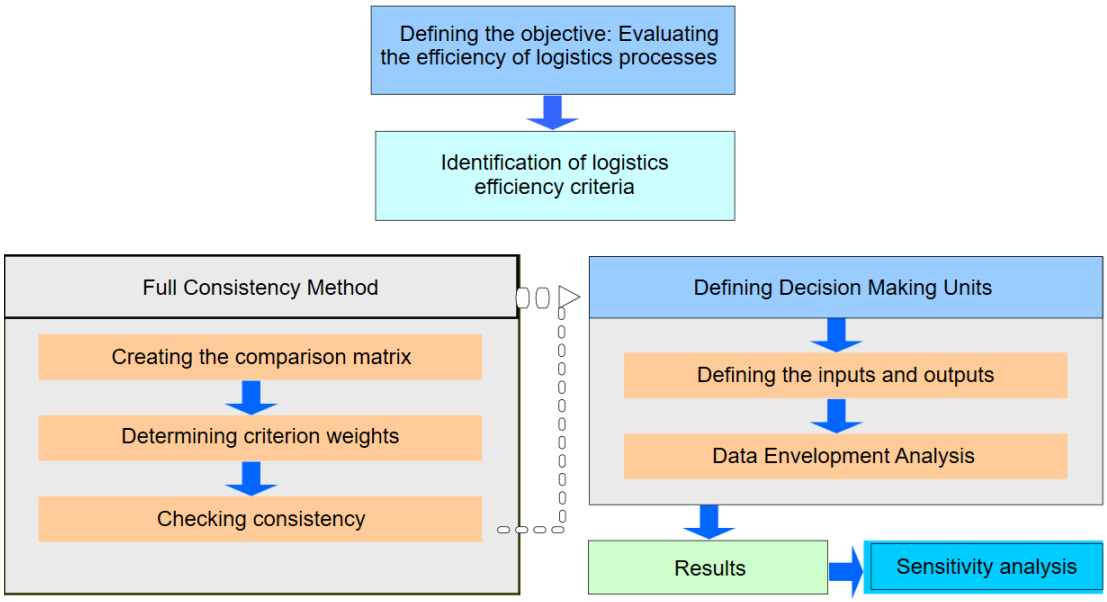


Figure 1. Methodological framework of the integrated FUCOM–DEA approach

Note: FUCOM: Full Consistency Method; DEA: Data Envelopment Analysis.

3.2 Full Consistency Method

The FUCOM represents a method for determining criteria weighting coefficients based on the principle of full consistency of expert judgments [8].

In this study, the FUCOM method was applied to determine the weights of input criteria related to logistics efficiency: material receiving time (C1), warehousing time (C2), and internal distribution time (C3). The obtained weights were used to define the AR constraints within the DEA model, while flexibility in determining DEA weights was retained for the output indicators.

The application of the FUCOM method involves three main steps. In the first step, the criteria are ranked according to their importance, from the most significant to the least significant:

$$C_{j(1)} > C_{j(2)} > \cdots > C_{j(k)} \tag{1}$$

where, k represents the rank of the observed criterion.

In the second step, pairwise comparisons of adjacent ranked criteria are performed, and their comparative priorities are determined: $\left(\varphi_{k/(k+1)}\right) = 1, 2, \ldots, n$, where, k represents the rank of the evaluation criterion. The comparative importance of the evaluation criteria $\varphi_{k/(k+1)}$ represents the advantage of the criterion ranked as $C_{j(k)}$ represents the advantage of the criterion ranked as $C_{j(k+1)}$. In this way, the vectors of comparative importance of evaluation criteria are obtained, as expressed in Eq. (2):

$$\Phi = \left(\varphi_{1/2}, \varphi_{2/3}, \ldots, \varphi_{k/(k+1)}\right) \tag{2}$$

In the third step, the final values of the weighting coefficients are determined. The ratio between the weights of adjacent criteria should correspond to the previously defined comparative priorities:

$$\frac{w_k}{w_{k+1}} = \varphi_{k/(k+1)} \tag{3}$$

In addition, the condition of mathematical transitivity must be satisfied:

$$\frac{w_k}{w_{k+2}} = \varphi_{k/(k+1)} \times \varphi_{(k+1)/(k+2)} \tag{4}$$

The final values of the weighting coefficients are determined by minimizing the deviation from full consistency:

$$\min \chi \tag{5}$$

s.t.:

$$\left| \frac{w_{j(k)}}{w_{j(k+1)}} - \varphi_{k/(k+1)} \right| = \chi, \forall j \tag{6}$$

$$\left| \frac{w_{j(k)}}{w_{j(k+1)}} - \varphi_{k/(k+1)} \times \varphi_{(k+1)/(k+2)} \right| = \chi, \forall j \tag{7}$$

$$\sum_{j=1}^{n} w_j = 1 \tag{8}$$

$$w_j \geq 0, \forall j \tag{9}$$

By solving the defined model, the final values of the weighting coefficients are obtained:

$$(w_1, w_2, \ldots, w_n)^{\mathrm{T}} \tag{10}$$

as well as the value of deviation from full consistency (χ). where, the superscript T denotes transposition. A lower value of (χ) indicates a higher level of consistency in expert judgments, while $\chi = 0$ denotes full consistency [8]. The obtained FUCOM weighting coefficients were used to define the AR constraints for input weights within the DEA model.

For the purpose of applying the FUCOM method, an expert evaluation of the criteria was conducted. The study involved five experts in the fields of logistics and industrial processes, with an average professional experience of more than 10 years. Expert evaluations were collected through a structured questionnaire, within which the experts first ranked the criteria according to their importance for evaluating the efficiency of logistics processes in the process industry and subsequently determined the comparative importance of adjacent ranked criteria, which represents the basis of the FUCOM method. The final values of comparative importance were obtained as the average values of expert assessments and were used to calculate the criteria weighting coefficients. The obtained value of the consistency degree confirmed the mutual agreement of the expert judgments.

3.3 Data Envelopment Analysis Method

DEA is a non-parametric method based on linear programming that enables the evaluation of the relative efficiency of DMUs based on multiple input and output indicators. The Charnes Cooper, and Rhodes (CCR) method was developed by Charnes et al. [21], while Banker et al. [22] extended the DEA approach by introducing the assumption of variable returns to scale (Banker, Charnes, and Cooper—BCC method).

In this study, an input-oriented CCR model was applied, based on the assumption of constant returns to scale. The model enables the evaluation of the relative efficiency of the analyzed DMUs in relation to the constructed efficiency frontier.

Since FUCOM weights are integrated into the DEA model through AR constraints, the multiplier form of the CCR model was used:

$$\max \sum_{r=1}^{s} u_r y_{r0} \tag{11}$$

With restrictions:

$$\sum_{i=1} v_i x_{i0} = 1 \tag{12}$$

$$\sum_{r=1}^{s} u_r y_{rj} - \sum_{i=1}^{m} v_i x_{ij} \leq 0, j = 1, 2, 3, \ldots, n \tag{13}$$

$$u_r \geq \varepsilon, r = 1, 2, 3, \ldots, s \tag{14}$$

$$v_i \geq \varepsilon, i = 1, 2, 3, \ldots, m \tag{15}$$

where, x_{ij} represents the value of the i-th input for the j-th DMU, y_{rj} represents the value of the r-th output, v_i and u_r represent the input and the output weights, while m, s and n indicate the number of inputs, outputs, and analyzed DMUs. The value of ε represents a small positive number ($\varepsilon = 10^{-6}$), which prevents the assignment of zero weights. The optimal value of the objective function h_0 represents the relative efficiency score of the observed DMU and is within the following interval:

$$0 \leq h_0 \leq 1 \tag{16}$$

The value $h_0 = 1$ indicates a relatively efficient DMU, whereas $h_0 < 1$ indicates relative inefficiency and the existence of opportunities for improving the utilization of input resources.

In order to integrate the expert assessments obtained through the FUCOM method into the DEA model, the AR constraint approach, introduced into the DEA literature by Thompson et al. [23], was applied. This approach enables the incorporation of information on the relative importance of input criteria by restricting the ratios of DEA weights. In this way, the DEA model retains flexibility in weight selection while simultaneously ensuring consistency with the priorities defined by experts.

In this study, FUCOM weights were determined only for the input criteria, while the output weights were left unrestricted in accordance with the basic DEA methodology. Let w_i^F denote the normalized FUCOM weights of the input criteria. The ratio of DEA input weights is then constrained as follows:

$$\frac{v_i}{v_j} \in \left[\frac{w_i^F}{w_j^F}(1-\delta), \frac{w_i^F}{w_j^F}(1+\delta) \right], \forall i, j, i \neq j \tag{17}$$

where, δ represents the allowable deviation from the ratio of FUCOM weights. In this study, a value of δ = 0.1 was adopted, providing an appropriate balance between the flexibility of the DEA model and the preservation of expert-defined priorities, in accordance with the recommendations from the literature [23]. Based on the above, the final CCR multiplier DEA–AR model can be defined as follows:

$$\max \sum_{r=1}^{s} u_r y_{r0} \tag{18}$$

With restrictions:

$$\sum_{i=1}^{m} v_i x_{i0} = 1 \tag{19}$$

$$\sum_{r=1}^{s} u_r y_{rj} - \sum_{i=1}^{m} v_i x_{ij} \leq 0, j = 1, 2, 3, \ldots, n \tag{20}$$

$$\frac{v_i}{v_j} \leq \frac{w_i^F}{w_j^F}(1+\delta), \forall i, j, i \neq j \tag{21}$$

$$\frac{v_i}{v_j} \geq \frac{w_i^F}{w_j^F}(1-\delta), \forall i, j, i \neq j \tag{22}$$

$$v_i \geq \varepsilon, i = 1, 2, 3, \ldots, m \tag{23}$$

In this way, the DEA model allows for an objective assessment of the relative efficiency of logistics processes, while simultaneously taking into account expert preferences defined by the FUCOM method. Such an approach represents a methodological improvement over classic DEA models without weight restrictions, as it allows for the integration of subjective expert assessments into the optimization framework of the DEA model.

4 Case Study and Results

The case study was conducted at the Modriča Oil Refinery, as a representative example of the process industry in Bosnia and Herzegovina. This company is one of the most significant industrial companies in Bosnia and Herzegovina. It is specialized in the production of motor and industrial oils, lubricants and special fluids, with a developed production infrastructure and a long-standing business tradition. The Modriča Oil Refinery was founded in 1954, and during its development it transformed into a modern industrial system with a wide production portfolio and developed logistics capacities. Today it operates as a joint-stock company with an integrated organizational structure that includes production facilities, quality control laboratories, storage systems and logistics units for the distribution of products to the domestic and international markets. The company has modern technological capacities for processing and blending of products, as well as automated packaging and storage lines. The product portfolio includes more than 250 products that are placed on international markets, ensuring a stable market position and competitiveness. Objective of the case study is to analyze the efficiency of logistics activities related to the management of raw material flows and their supply to production facilities. The refinery logistics system is an integrated system for managing the flows of raw materials, intermediate products and finished products through technological and distribution phases. As shown in Figure 2, the logistics process includes the procurement and receipt of raw materials, storage in tank capacities, internal distribution to production facilities, the production process, quality control, packaging and storage of finished products and distribution to the domestic and international markets.

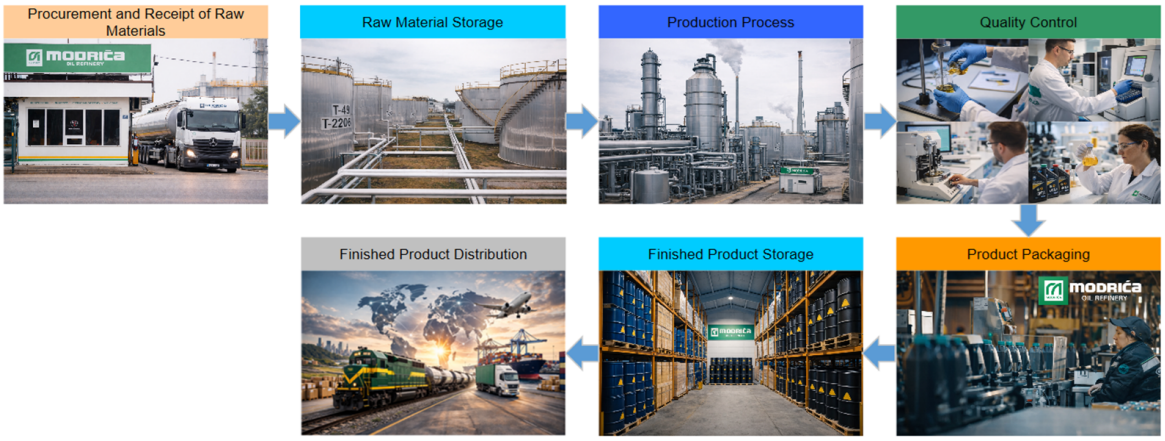


Figure 2. Logistics process at the Modriča Oil Refinery

The process begins with the receipt of raw materials by road transport. Afterward, the raw materials are stored in tank systems that ensure continuity of supply. From the warehouse, the raw materials are transported via pipelines to the production facilities, where the mixing, stabilization and homogenization processes take place. In parallel with production, laboratory quality control is carried out by testing key parameters. After quality verification, the products are redirected to automated packaging lines, where filling, sealing, and labeling of packaging are carried out. Finished products are stored and distributed to the domestic and international markets.

Although the refinery logistics-production system includes several interconnected phases, the analysis in this study focuses on the logistics processes related to the management of incoming raw material flows. In accordance with the research objective, the DEA analysis includes three key logistics phases: 1) raw material receipt, 2) raw material storage, and 3) internal distribution to production facilities. These phases represent the basic functional units of the logistics system and are modeled as DMUs. This enables a detailed analysis of the efficiency of inbound logistics flows, along with the identification of potential sources of inefficiency and opportunities for improving logistics performance.

4.1 Data Collection and Preparation

The empirical analysis is based on operational logistics data related to the receipt, storage and internal distribution of raw materials at the Modriča Oil Refinery for a period of one year (January–December). Based on the available data, representative time parameters of logistics processes were defined. The average time for receiving base oils is approximately 75 minutes, while receiving packaging takes about 45 minutes, and for additives approximately 28 minutes. The storage time of materials varies depending on the type of raw material and production dynamics. Base oils and additives are stored for a period of one to ninety days, while the storage period of packaging is shorter and ranges from one to thirty days. Internal distribution of materials to production facilities is organized through various logistics systems. Base oils are transported via a pipeline network directly to production lines, while packaging and additives are distributed by internal transport vehicles from the warehouse to production.

For the purposes of DEA analysis, DMUs are defined as logistical material flows observed on a monthly basis. In this way, a set of 36 DMU units was formed in total (12 months × 3 material types), as presented in Table 1.

The input variables in the model include:

• Material receipt time (minutes): the total time from vehicle arrival to the completion of unloading and documentation procedures.

• Storage time (days): the average time materials remain in storage before being used in production.

• Internal distribution time (minutes): the time required to transport materials from the storage facility to the point of use in production.

The output variables include:

• Processed material quantity (tons): the total quantity of material that passed through the logistics process during the observed month.

• Supply reliability (%): the percentage of successfully fulfilled production requests relative to the total number of received requests during the observed month, defined as:

$$\text{Realiability } = \frac{\sum_{k=1}^{K} I\left(\text{idelivery }_k \leq \text{ time }_k\right)}{K} \times 100\%$$

where, K represents the total number of supply requests during the observed period, and I is an indicator function that takes the value 1 if the delivery is completed on time. This metric was chosen because it directly reflects the ability of the logistics system to meet the operational needs of production, which is one of the key objectives of supply chain management in the process industry. In this way, a consistent set of input–output parameters was established, that allows a reliable assessment of the efficiency of logistics processes using the DEA model.

4.2 Results of the Full Consistency Method

In order to determine the relative importance of the criteria of logistics efficiency, the FUCOM method was applied. Five experts (E1–E5) in the field of logistics and industrial process management with an average work experience of more than ten years participated in the research. The experts first ranked the criteria (C1–C3) according to their importance for assessing the efficiency of logistics processes in the process industry. Based on the expert assessment, the results of the weighting coefficient values are shown in Table 2.

The results show that the criterion of material receipt time has the greatest impact on logistics process efficiency, with a weight of 0.413. This result indicates that timely receipt of raw materials is a key prerequisite for the smooth operation of the production process and the efficient functioning of the logistics system. The storage time criterion takes second place, with a weight of 0.344, which confirms the importance of inventory management and optimization of storage capacities. Internal distribution time has a weight of 0.243, placing it third in terms of importance, while still representing an important factor in overall logistics efficiency. The value of the deviation from full consistency

parameter was χ = 0.000, confirming complete consistency of expert evaluations and a high level of reliability of the obtained weight coefficients.

Table 1. Input and output data for DEA analysis

DMU	Month	Material Type	Receipt (min)	Storage (days)	Distribution (min)	Quantity (tons)	Reliability
DMU1	January	Base oils	74	46	10	1205	0.960
DMU2	January	Packaging	44	20	15	900	0.930
DMU3	January	Additives	27	30	20	800	0.920
DMU4	February	Base oils	76	49	11	1220	0.964
DMU5	February	Packaging	46	22	16	910	0.934
DMU6	February	Additives	28	32	19	810	0.925
DMU7	March	Base oils	73	43	9	1240	0.968
DMU8	March	Packaging	43	19	14	920	0.940
DMU9	March	Additives	26	29	19	820	0.930
DMU10	April	Base oils	77	41	10	1265	0.972
DMU11	April	Packaging	45	18	14	930	0.945
DMU12	April	Additives	29	28	18	825	0.934
DMU13	May	Base oils	75	45	10	1230	0.966
DMU14	May	Packaging	47	21	15	915	0.938
DMU15	May	Additives	30	31	20	815	0.929
DMU16	June	Base oils	72	47	9	1215	0.963
DMU17	June	Packaging	44	23	16	905	0.933
DMU18	June	Additives	27	33	21	805	0.923
DMU19	July	Base oils	78	44	11	1275	0.974
DMU20	July	Packaging	46	20	15	935	0.946
DMU21	July	Additives	31	30	19	830	0.936
DMU22	August	Base oils	74	42	10	1250	0.970
DMU23	August	Packaging	45	19	14	925	0.942
DMU24	August	Additives	28	29	18	822	0.931
DMU25	September	Base oils	79	48	12	1280	0.976
DMU26	September	Packaging	48	24	17	940	0.948
DMU27	September	Additives	30	34	20	835	0.938
DMU28	October	Base oils	75	46	10	1245	0.968
DMU29	October	Packaging	44	21	15	918	0.939
DMU30	October	Additives	29	31	19	818	0.930
DMU31	November	Base oils	73	43	9	1225	0.965
DMU32	November	Packaging	43	18	14	908	0.935
DMU33	November	Additives	26	28	18	808	0.924
DMU34	December	Base oils	76	45	10	1260	0.971
DMU35	December	Packaging	46	20	16	930	0.944
DMU36	December	Additives	28	30	19	824	0.932

Note: DEA: Data Envelopment Analysis; DMU: decision-making units.

Table 2. Values of the weight coefficients

Criteria ID	Criterion	E1	E2	E3	E4	E5	w_j
C1	Material time	0.413	0.462	0.450	0.333	0.406	0.413
C2	Storage time	0.344	0.308	0.300	0.400	0.369	0.344
C3	Internal distribution time	0.243	0.231	0.250	0.267	0.225	0.243

Note: E1–E5, five experts; C1–C3, three ranked criteria. w_j represents the final weight coefficient of criterion j.

4.3 Results of the Data Envelopment Analysis Method

After determining the weight coefficients of the input criteria using the FUCOM method, the relative efficiency of logistics units was assessed using an input-oriented CCR DEA model with AR constraints. In this way, expert-defined priorities were included in the DEA optimization model through constraints on the weight ratios of the input criteria, which ensured greater reality of the obtained results compared to the classic DEA model.

The values shown in Table 3 represent the limits of the DEA weight ratios defined based on the FUCOM weight coefficients and the adopted level of deviation (δ = 0.10). These constraints allow the optimization procedure of the DEA model to be aligned with expert assessments of the importance of the input criteria, while preserving the basic principles of the DEA methodology. It can be seen that all DEA weight ratios are defined in relatively narrow intervals, which allows the model to maintain sufficient flexibility in optimization while preventing unrealistic

deviations from expert-determined priorities. After implementing the AR constraints, an assessment of the relative efficiency of all analyzed logistics flows was performed, and the results are shown in Table 4.

Table 3. AR constraints on input weight ratios for $\delta = 0.10$

DEA Weight Ratio	Lower Limit	Upper Limit
v_1/v_2	1.081	1.321
v_1/v_3	1.530	1.870
v_2/v_3	1.274	1.557

Note: DEA: Data Envelopment Analysis; AR: Assurance Region. v_1–v_3 denote the input weights.

Table 4. DEA–AR efficiency scores by material flow

Material Flow	Average Efficiency	Standard Deviation	Minimum	Maximum
Base oils	0.771	0.022	0.728	0.800
Packaging	0.947	0.041	0.880	1.000
Additives	0.937	0.039	0.884	1.000

Note: DEA: Data Envelopment Analysis; AR: Assurance Region.

The results of the DEA–AR analysis shown in Table 4 show that the average efficiency values differ between the analyzed logistics flows. The highest average efficiency is achieved by the logistics flow of packaging with a value of 0.947, followed by the logistics flow of additives with an average efficiency of 0.937, while the logistics flow of base oils achieves the lowest average efficiency of 0.771. Based on the average efficiency values, the final ranking of logistics flows can be determined: Packaging $>$ Additives $>$ Base oils. The obtained results confirm that the application of the DEA–AR model allows for the clear identification of logistics flows with the greatest potential for improvement, which is an important basis for making decisions on the optimization of logistics processes in the process industry.

The high efficiency of the logistics flow of packaging primarily results from the specifics of this process. First of all, the receipt and manipulation of packaging require a smaller number of logistics operations compared to base oils, while the average storage time is significantly shorter due to the higher frequency of consumption in the production process. In addition, the internal distribution of packaging is carried out through standardized transport routes and relatively simple handling operations, which contributes to a more stable production supply dynamic. Such an organization of the logistics process results in lower variability of operating times and a more favorable ratio between resources used and outputs achieved, which the DEA model recognizes as a higher level of efficiency.

Looking at the standard deviation, it can be concluded that the logistics flow of base oils shows the lowest variability of results (0.022), which indicates stable functioning of the process throughout the entire observed period. However, process stability does not necessarily imply high efficiency. The DEA model estimates the ratio between used input resources and achieved outputs, which is why a stable process can be assessed as less efficient if it uses relatively more resources to achieve similar outputs.

The highest maximum efficiency (1.000) was recorded in the logistics flow of packaging, with two DMU units identified as fully efficient. At the same time, the minimum efficiency value is 0.880, which indicates that even the least efficient months achieve a relatively high level of performance. These results confirm that the packaging logistics process is organized in a way that enables rational use of available resources with a high level of reliability of production supply.

Similar results were obtained for the additive logistics flow. The average efficiency of 0.937 indicates that this logistics flow achieves very good performance, although some months showed certain deviations due to increased storage time and internal distribution. The maximum efficiency of 1.000 was achieved in November, while the minimum value was 0.884. The slightly lower efficiency compared to packaging materials is a consequence of the greater sensitivity of the logistics process to changes in production dynamics and higher variations in the storage time of individual additives, which are used in different formulations and in smaller quantities.

Unlike the previous two logistics flows, the base oil logistics flow achieved an average efficiency of 0.771. This result is primarily a consequence of higher values of the input criteria, particularly receipt time and storage time. Although base oils achieve the highest processed material quantities and a very high level of supply reliability, the increased input resource values reduce the relative DEA efficiency. In other words, the achieved outputs were not sufficient to compensate for the higher input requirements compared to the other logistics flows. Additionally, handling base oils requires the use of tank storage capacities, pipeline infrastructure, and strictly defined receipt and storage procedures, which increases the duration of certain logistics activities. Although such procedures are necessary from the point of view of product safety and quality, they also reduce the relative technical efficiency

measured by the DEA model. This indicates that improving the logistics process of base oils should primarily focus on reducing the time of material receipt and optimizing the storage period, since these criteria have the greatest impact on the relative efficiency of the logistics system. The analysis of individual DMUs has been shown in Table 5.

Table 5. Individual efficiency scores of DMUs

DMU	Month	Material Type	DEA–AR Efficiency
DMU1	January	Base oils	0.750
DMU2	January	Packaging	0.945
DMU3	January	Additives	0.937
DMU4	February	Base oils	0.728
DMU5	February	Packaging	0.901
DMU6	February	Additives	0.914
DMU7	March	Base oils	0.799
DMU8	March	Packaging	1.000
DMU9	March	Additives	0.992
DMU10	April	Base oils	0.794
DMU11	April	Packaging	0.995
DMU12	April	Additives	0.971
DMU13	May	Base oils	0.764
DMU14	May	Packaging	0.910
DMU15	May	Additives	0.894
DMU16	June	Base oils	0.768
DMU17	June	Packaging	0.912
DMU18	June	Additives	0.898
DMU19	July	Base oils	0.773
DMU20	July	Packaging	0.954
DMU21	July	Additives	0.915
DMU22	August	Base oils	0.800
DMU23	August	Packaging	0.977
DMU24	August	Additives	0.971
DMU25	September	Base oils	0.746
DMU26	September	Packaging	0.880
DMU27	September	Additives	0.884
DMU28	October	Base oils	0.768
DMU29	October	Packaging	0.953
DMU30	October	Additives	0.919
DMU31	November	Base oils	0.789
DMU32	November	Packaging	1.000
DMU33	November	Additives	1.000
DMU34	December	Base oils	0.776
DMU35	December	Packaging	0.941
DMU36	December	Additives	0.952

Note: DEA: Data Envelopment Analysis; DMUs: decision-making units; AR: Assurance Region.

The analysis of individual DMUs shows that the best results were achieved in March and November for the logistics flow of packaging, as well as in November for the logistics flow of additives, where full DEA efficiency was recorded. In contrast, the lowest efficiency was recorded for base oils in February (0.728), which indicates that this period represents the greatest potential for optimizing logistics activities. Observed by months, it can be seen that the logistics flows of packaging and additives achieve high DEA efficiency values for most of the year, which indicates a stable level of logistics activity organization. On the other hand, for base oils, there are more pronounced deviations between individual months, especially in February and September, which may indicate increased load on receiving capacities, longer storage times or changes in production dynamics. Such results confirm that DEA analysis allows the identification of time periods in which there is the greatest potential for improving logistics processes.

The descriptive statistics presented in Table 6 further confirm the previous conclusions. The average efficiency of all analyzed DMU units is 0.885, while the standard deviation of 0.089 indicates a moderate dispersion of results. The maximum achieved efficiency value is 1.000, while the minimum value is 0.728, confirming the existence of a certain potential for improving logistics processes.

Table 6. Descriptive statistics of DEA–AR efficiency

Indicator	Value
Maximum efficiency	1.000
Minimum efficiency	0.728
Average efficiency	0.885
Standard deviation	0.089

Note: DEA: Data Envelopment Analysis; AR: Assurance Region.

The results of the descriptive statistics confirm that the majority of analyzed logistics units achieve a relatively high level of efficiency, while the moderate value of the standard deviation indicates the existence of differences among individual logistics flows that justify a more detailed analysis of their performance. The obtained results provide the basis for forming a composite FUCOM–DEA score, which further evaluates logistics flows by integrating the results of the DEA analysis and the weighted criteria obtained using the FUCOM method.

4.4 Composite Full Consistency Method–Data Envelopment Analysis Score

For the purpose of additional evaluation of logistics flows, a composite FUCOM–DEA score was developed. This score combines three elements: the average DEA–AR efficiency, a weighted input score obtained by applying FUCOM weights, and an output score based on the amount of processed material and supply reliability. It is important to emphasize that FUCOM weights were used exclusively for weighting the input criteria, in accordance with the methodological framework of the research. The output score is not determined by applying the FUCOM method. It is obtained by normalizing the output indicators (quantity of processed material and supply reliability), which allows them to be combined with the results of the DEA analysis within the framework of a composite FUCOM–DEA score.

Input criteria are treated as cost criteria, whereby lower values of reception, storage and internal distribution times represent more favorable logistics performance. In contrast, the output indicators were normalized as benefit criteria, where higher values of processed material quantity and supply reliability indicate better performance of the logistics process. Based on the composite FUCOM–DEA score (Table 7), the same final ranking was obtained as in the DEA-AR analysis: Packaging materials $>$ Additives $>$ Base oils.

Table 7. Composite FUCOM–DEA score

Material Flow	DEA–AR Efficiency	Input Score	Output Score	Composite FUCOM–DEA Score
Packaging	0.947	0.765	0.855	0.879
Additives	0.937	0.772	0.809	0.864
Base oils	0.771	0.555	1.000	0.774

Note: FUCOM: Full Consistency Method; DEA: Data Envelopment Analysis; AR: Assurance Region.

The results show that the packaging material logistics flow achieves the highest composite score of 0.879. This result confirms that packaging materials have the most favorable relationship between the utilized input resources and the achieved output performance. Although the quantities of packaging materials are not the highest, shorter receipt and storage times significantly contribute to better overall efficiency.

The additive logistics flow achieves a composite score of 0.864, placing it second in the ranking. Additives have a very favorable input score, but a slightly lower output score compared to base oils, primarily due to a lower processed material quantity and lower supply reliability.

Base oils achieve the highest output score because they have the largest processed material quantities and the highest supply reliability. However, due to significantly higher receipt and storage times, their input score is the lowest, which affects the overall composite result. Therefore, despite their high output performance, base oils rank third in the final ranking.

The composite FUCOM–DEA score confirms the consistency of the previously obtained results and demonstrates that the final ranking of logistics flows is not determined solely by DEA efficiency, but also by the broader relationship between input requirements and output performance.

5 Sensitivity Analysis

In order to verify the robustness of the proposed integrated FUCOM–DEA methodological framework, a sensitivity analysis of the results was conducted at two levels: (1) the stability of the criterion weighting system and (2) the stability of the efficiency of logistics flows obtained using the DEA model with AR constraints.

5.1 Sensitivity Analysis of the Full Consistency Method

In order to verify the stability of the weighting system obtained using the FUCOM method, a sensitivity analysis (Table 8) was conducted by varying individual criterion weights. The initial weight values are: C1—material receipt time 0.413, C2—storage time 0.344, and C3—nternal distribution time 0.243. In each scenario, the weight of one criterion was changed, after which the weight vector was re-normalized so that the sum of the weights remained equal to 1. In order to examine the stability of the criteria ranking, a sensitivity analysis was conducted by varying the weights of the criteria in the range of ±10% and ±20%, with mandatory re-normalization of the weight vector.

Table 8. Sensitivity analysis of FUCOM criterion weights

Scenario	C1	C2	C3	Rank
Baseline model	0.413	0.344	0.243	C1 > C2
C1 -10%	0.388	0.359	0.253	C1 > C2
C1 -20%	0.360	0.375	0.265	C2 > C3
C1 +10%	0.436	0.330	0.233	C1 > C2
C1 +20%	0.458	0.318	0.224	C1 > C2
C2 -10%	0.428	0.321	0.252	C1 > C2
C2 -20%	0.444	0.296	0.261	C1 > C2
C2 +10%	0.399	0.366	0.235	C1 > C2
C2 +20%	0.386	0.386	0.227	C1 > C2
C3 -10%	0.423	0.352	0.224	C1 > C2
C3 -20%	0.435	0.362	0.205	C1 > C2
C3 +10%	0.403	0.336	0.261	C1 > C2
C3 +20%	0.394	0.328	0.278	C1 > C2

Note: FUCOM: Full Consistency Method; C1–C3, three ranked criteria.

The results of the sensitivity analysis show that the weighting system is relatively stable. In most scenarios, the basic ranking of the criteria remains unchanged, i.e. C1 > C2 > C3. The change in rank occurs only in the scenario of reducing the weight of criterion C1 by 20%, when the storage time criterion takes first place. Also, when the weight of C2 is increased by 20%, the importance of criteria C1 and C2 is equalized. These results indicate that the time of receipt of materials and the time of storage are the most sensitive criteria in the model, while the criterion of internal distribution time remains stable in third place in all analyzed scenarios.

5.2 Sensitivity Analysis of the Data Envelopment Analysis-Assurance Region Model

The sensitivity analysis of the DEA–AR model (Table 9) was conducted by changing the parameter δ, which defines the permissible deviation of DEA weight ratios from the FUCOM weight ratios. In the baseline model, the value of $\delta = 0.10$ was used, while additional scenarios with $\delta = 0.20$, $\delta = 0.30$ and $\delta = 0.40$ were analyzed.

Table 9. Sensitivity analysis of the DEA–AR model

Scenario	Base Oils	Packaging	Additives	Rank
$\delta = 0.10$	0.771	0.947	0.937	Packaging > Additives > Base oils
$\delta = 0.20$	0.779	0.948	0.938	Packaging > Additives > Base oils
$\delta = 0.30$	0.788	0.948	0.939	Packaging > Additives > Base oils
$\delta = 0.40$	0.801	0.948	0.940	Packaging > Additives > Base oils

Note: DEA: Data Envelopment Analysis; AR: Assurance Region.

The results show that changing the parameter δ does not affect the final ranking of logistics flows. In all scenarios, the highest average efficiency is achieved by the packaging flow, followed by the additives flow, while the base oil flow remains ranked third. This confirms the stability of the DEA–AR model and shows that the conclusions obtained are not the result of only one selected value of the parameter δ.

Increasing the value of δ leads to a smaller increase in the average efficiency of base oils, from 0.771 to 0.801, which indicates that this logistics flow has a certain benefit from greater flexibility of weights in the DEA model. However, even with a larger permissible deviation of the weight ratio, the final ranking does not change. For packaging and additives, the changes are minimal, which confirms that their results are stable and less dependent on the width of the AR restrictions.

Based on the analysis conducted, it can be concluded that the proposed FUCOM–DEA model has satisfactory robustness. FUCOM weights show a stable ranking of criteria, while DEA–AR results remain stable when changing

the parameter δ. This confirms the reliability of the model for assessing the efficiency of logistics processes in the analyzed industrial system.

6 Discussion

The integration of FUCOM and DEA provided a structured framework for assessing the efficiency of logistics processes in the process industry. FUCOM was used to determine the weights of the input criteria from expert judgments, while the DEA–AR model measured the relative efficiency of the logistics flows under AR constraints. Incorporating the FUCOM-derived priorities into the DEA model restricted its freedom to assign disproportionately small or zero weights to operationally important inputs. This addressed a recognized limitation of conventional DEA and aligned the efficiency assessment more closely with the operating priorities of the refinery.

The DEA results showed that the packaging material flow achieved the highest average efficiency, followed by the additive flow, whereas the base-oil flow recorded the lowest average efficiency. These results indicate that logistics efficiency was not determined solely by the quantity of material processed or the reliability of supply. It also depended on the relationship between the logistics resources consumed and the outputs produced. This interpretation is consistent with the principles of DEA, according to which relative efficiency is evaluated through the relationship between multiple inputs and outputs rather than through the absolute value of any single indicator.

The efficiency of the packaging material flow may be explained by the organizational and technological characteristics of this process. The receipt of packaging materials generally involved shorter handling times, shorter storage periods, and standardized internal distribution procedures with relatively few handling activities. These characteristics produced a favorable relationship between the three input variables—receipt time, storage time, and internal distribution time—and the two outputs—processed material quantity and supply reliability. The comparatively limited variation in its operating times may also explain why the packaging material flow maintained high DEA efficiency during most of the observed period.

The additive flow achieved slightly lower average efficiency than the packaging material flow but still demonstrated comparatively strong logistics performance. A possible explanation is the greater variation in storage and internal distribution times resulting from differences in production formulations and changes in the consumption of individual additives. Despite this variability, relatively short receipt times and the organization of the storage activities allowed the additive flow to maintain a favorable relationship between its logistics inputs and outputs.

The base-oil flow recorded the lowest DEA efficiency even though it handled the largest quantities of material and maintained a high level of supply reliability. This result should not be interpreted as direct evidence of poor logistics management. It may instead reflect the specific technological and regulatory requirements associated with base-oil handling. The receipt process includes several control activities, tank storage, pipeline transportation, and compliance with strict safety and quality-control procedures. These requirements may increase receipt and storage times, which are treated as inputs in the DEA model and may consequently reduce relative technical efficiency.

The interpretation of these comparisons requires attention to the homogeneity of the DMUs. Although the base-oil, packaging material, and additive flows differ in their physical characteristics and handling requirements, all 36 DMUs performed the same general logistics function, consisting of material receipt, storage, and internal distribution. They were also evaluated using the same definitions of the three inputs and two outputs. These common features allowed them to be assessed within a shared DEA framework. Nevertheless, the technological differences among the three material flows may affect their operating times and should be considered when interpreting the common efficiency frontier.

The FUCOM results identified material receipt time as the most important input criterion of logistics efficiency. This finding is consistent with the operating conditions of the process industry, where delays in receiving raw materials may affect production continuity, storage capacity utilization, and logistics costs. Storage time was ranked second, indicating the importance of inventory management and the timely movement of materials through the logistics system. Internal distribution time received the lowest relative weight among the three inputs, although it remained an integral part of the efficiency assessment. The weighting procedure followed the consistency principles of FUCOM [8], allowing the experts' comparative judgments to be converted into a consistent set of criterion weights.

An important methodological feature of this study is the incorporation of FUCOM weights into the DEA model through AR constraints. Previous applications have frequently used FUCOM as a separate criterion-weighting method or combined it with ranking methods such as MARCOS, MABAC, and CoCoSo. In the present study, the information obtained from FUCOM was used to define restrictions on the ratios of the DEA weights. The resulting model retained the quantitative efficiency-measurement function of DEA while preventing the optimized weights from departing excessively from the operational priorities expressed by the experts. The framework therefore connected expert judgment and operational data within the same efficiency model rather than treating them as two unrelated analytical stages.

The findings are generally consistent with earlier research showing that MCDM methods and DEA can be combined when the assessment of a complex system requires both preference information and operational

performance data. Abdullah et al. [10] and Khosravi et al. [11] applied FUCOM–MARCOS frameworks to healthcare-system performance. Mitrović Simić et al. [14] developed hybrid DEA models for evaluating transportation infrastructure safety. Those studies focused primarily on healthcare systems, sustainability, transportation safety, suppliers, or external logistics providers. The present research extends this methodological stream to internal logistics in the process industry by jointly examining material receipt, storage, and internal distribution within an oil refinery.

The sensitivity assessment considered changes in the FUCOM weights as well as changes in the allowable weight ratios within the DEA–AR model. In the first part, the weights of the input criteria were varied by ±10% and ±20%. These changes did not produce material alterations in the final ordering of the logistics flows. The largest variation was associated with material receipt time, which had received the highest FUCOM weight. Changes in storage time and internal distribution time produced smaller variations in the efficiency results. The model was therefore more responsive to changes in its most highly weighted input, as would be expected from the FUCOM results.

In the second part of the sensitivity assessment, the value of the δ parameter defining the allowable deviation of the weight ratios within the AR constraints was changed. The final ordering of the three logistics flows remained unchanged across the examined δ scenarios. Taken together, these findings show that the main efficiency pattern was stable under the tested changes in expert-derived weights and AR specifications. This stability does not eliminate the subjective element of the FUCOM assessment, but it indicates that moderate changes in the input assumptions did not determine the principal empirical findings.

From a managerial perspective, the results identify several priorities for logistics improvement. For the base-oil flow, the greatest scope for improvement concerns the duration and organization of material receipt, delivery planning, coordination between procurement schedules and production plans, and inventory management. However, any attempt to reduce receipt and storage times must account for the safety, quality-control, and technological requirements associated with tank storage and pipeline handling. A lower DEA score should therefore be used as a starting point for process examination rather than as sufficient evidence that necessary control activities should be removed.

The packaging material and additive flows already recorded relatively high levels of efficiency. Further improvement in these flows may be sought through additional automation of internal handling operations, closer scheduling of deliveries, and reductions in avoidable internal distribution time. The operational value of these measures should be assessed against their implementation costs because a high DEA score does not by itself establish the economic justification for additional investment.

The study also has several limitations. The analysis was based on one company and one year of operational data, which restricts the generalization of the findings beyond the examined refinery. The model included three input and two output variables and did not explicitly account for logistics costs, energy consumption, emissions, storage capacity utilization, or service-quality differences among the material flows. The three categories of material were assessed within a common frontier, although their technological and handling requirements were not identical. Future research could examine a larger number of companies from different branches of the process industry, use longer time series, and incorporate economic, energy, environmental, and capacity-related indicators.

7 Conclusion

Efficient logistics management is essential to production continuity, resource utilization, and competitiveness in the process industry. Because internal logistics involves several interconnected inputs and outputs, isolated operational or financial indicators cannot provide a complete assessment of process efficiency. This study developed an integrated FUCOM–DEA framework that combined structured expert judgments regarding criterion importance with an operational data-based assessment of relative efficiency.

In the first stage, FUCOM was used to determine the weights of three logistics input criteria: material receipt time, storage time, and internal distribution time. Material receipt time received the highest weight, followed by storage time and internal distribution time. The FUCOM-derived priorities were subsequently used to define AR constraints in the DEA model. This procedure incorporated expert information into the DEA weight-optimization process while retaining the basic structure of relative efficiency measurement.

The empirical analysis used operational data from the Modriča Oil Refinery. A total of 36 DMUs were defined, representing monthly observations of the base-oil, packaging material, and additive logistics flows. All DMUs performed the general functions of receipt, storage, and internal distribution and were evaluated using three input and two output variables. An input-oriented CCR DEA model with AR constraints was then applied to measure their relative efficiency.

The packaging material flow achieved the highest average efficiency, followed by the additive flow, whereas the base-oil flow recorded the lowest average efficiency. Although the base-oil flow handled the largest quantities and maintained high supply reliability, its longer receipt and storage times reduced its relative position within the DEA model. These longer operating times may be associated with tank storage, pipeline transportation, and the additional safety and quality-control procedures required for base oils. The findings therefore show that high throughput and

reliable supply do not necessarily correspond to high technical efficiency when they require comparatively large logistics inputs.

The sensitivity assessment examined two sources of variation. First, changes of ±10% and ±20% were introduced into the FUCOM weights. These scenarios did not materially change the final ordering of the logistics flows, although changes in the weight of material receipt time produced the largest variation in the efficiency results. Second, the δ parameter governing allowable weight-ratio deviations within the AR constraints was varied. The final ordering of the logistics flows remained unchanged across the tested δ values. The principal findings were therefore stable under the sensitivity scenarios considered in the study.

The methodological contribution lies in integrating FUCOM-derived priorities directly into DEA through AR constraints. Rather than using FUCOM only to rank criteria before a separate efficiency analysis, the proposed framework used the criterion weights to regulate DEA weight flexibility. It thus provided a coherent link between expert-defined operational priorities and the observed input–output performance of the logistics processes. The application to internal refinery logistics also extends previous FUCOM–DEA research, which has predominantly addressed suppliers, transport systems, logistics providers, infrastructure safety, and warehouse equipment.

The practical contribution concerns the identification of logistics activities requiring managerial attention. The results direct attention to receipt and storage times in the base-oil flow and provide a basis for reviewing delivery planning, procurement–production coordination, inventory management, and the organization of internal distribution. The model may also be used for periodic performance monitoring, provided that the input and output variables and the AR restrictions are reviewed when operating conditions or managerial priorities change. With appropriate adaptation of the variables, the framework may be examined in chemical, food, pharmaceutical, petrochemical, and other process–industry settings.

The findings should nevertheless be interpreted within the boundaries of the case study. The analysis covered one refinery and one year of data, while the DEA specification contained three inputs and two outputs. Logistics costs, energy use, CO_2 emissions, storage capacity utilization, and other sustainability indicators were not included. Future studies should test the framework across several companies and longer periods, incorporate additional economic and environmental variables, and compare the CCR results with other approaches. Such extensions would clarify the effects of scale, input slacks, and relationships among individual stages of the internal logistics process.

Overall, the integrated FUCOM–DEA framework provides a structured means of evaluating internal logistics efficiency in the process industry. Combining consistent expert-based weighting with DEA–AR analysis supports the identification of inefficient logistics activities and the setting of operational improvement priorities. Within the tested case and sensitivity scenarios, the framework produced a stable efficiency ordering and offered decision-relevant information for logistics management.

Author Contributions

Conceptualization, M.B. and D.B.; methodology, M.B.; validation, Ž.S.; formal analysis, M.B.; data curation, M.B.; writing—original draft preparation, M.B. and D.B.; writing—review and editing, Ž.S.; supervision, Ž.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] W. W. Cooper, L. M. Seiford, and J. Zhu, *Handbook on Data Envelopment Analysis, 2nd ed.* New York, NY, USA: Springer, 2011. https://doi.org/10.1007/978-1-4419-6151-8

[2] M. Toloo and S. Nalchigar, "A new DEA method for supplier selection in presence of both cardinal and ordinal data," *Expert Syst. Appl.*, vol. 38, no. 12, pp. 14 726–14 731, 2011. https://doi.org/10.1016/j.eswa.2011.05.008

[3] E. A. Adalı and A. T. Işık, "Integration of DEMATEL, ANP and DEA methods for third party logistics providers' selection," *Manag. Sci. Lett.*, vol. 6, no. 5, pp. 325–340, 2016. https://doi.org/10.5267/j.msl.2016.3.004

[4] A. Blagojević, S. Vesković, S. Kasalica, A. Gojić, and A. Allamani, "The application of the fuzzy AHP and DEA for measuring the efficiency of freight transport railway undertakings," *Oper. Res. Eng. Sci. Theory Appl.*, vol. 3, no. 2, pp. 1–23, 2020. https://doi.org/10.31181/oresta2003001b

[5] Ž. Stević, S. Miškić, D. Vojinović, E. Huskanović, M. Stanković, and D. Pamučar, "Development of a model for evaluating the efficiency of transport companies: PCA–DEA–MCDM model," *Axioms*, vol. 11, no. 3, p. 140, 2022. https://doi.org/10.3390/axioms11030140

[6] K. Tone, "A slacks-based measure of efficiency in data envelopment analysis," *Eur. J. Oper. Res.*, vol. 130, no. 3, pp. 498–509, 2001. https://doi.org/10.1016/S0377-2217(99)00407-5

[7] N. Chai, W. Zhou, and Z. Jiang, "Sustainable supplier selection using an intuitionistic and interval-valued fuzzy MCDM approach based on cumulative prospect theory," *Inf. Sci.*, vol. 626, pp. 710–737, 2023. https://doi.org/10.1016/j.ins.2023.01.070

[8] A. N. Algawbery, M. A. Shukri, and M. Q. Al-Matwakel, "Optimizing wind plant locations in Yemen: A geospatial model using full consistency method (FUCOM)," *Energy Rep.*, vol. 15, p. 109096, 2026. https://doi.org/10.1016/j.egyr.2026.109096

[9] D. Ç. Aka, "A framework for the selection of material handling equipment using fuzzy FUCOM: A case study in the automotive industry," *Verimlilik Derg.*, vol. 59, no. 3, pp. 513–530, 2025. https://doi.org/10.51551/verimlilik.1575633

[10] A. Abdullah, S. Ahmad, M. A. Athar, N. Rajpoot, and F. Talib, "Healthcare performance management using integrated FUCOM-MARCOS approach: The case of India," *Int. J. Health Plann. Manage.*, vol. 37, no. 5, pp. 2635–2668, 2022. https://doi.org/10.1002/hpm.3488

[11] M. Khosravi, A. Haqbin, Z. Zare, and P. Shojaei, "Selecting the most suitable organizational structure for hospitals: An integrated fuzzy FUCOM-MARCOS method," *Cost Eff. Resour. Alloc.*, vol. 20, no. 1, p. 29, 2022. https://doi.org/10.1186/s12962-022-00362-3

[12] M. Blagojević, D. Blagojević, A. Danilevičius, E. Krmac, and S. A. Biancardo, "Application of the FUCOM and MARCOS methods for selecting logistics service providers," *J. Intell. Manag. Decis.*, vol. 4, no. 4, pp. 288–297, 2025. https://doi.org/10.56578/jimd040404

[13] S. Starčević, N. Bojović, R. Junevičius, and V. Skrickij, "Analytical hierarchy process method and data envelopment analysis application in terrain vehicle selection," *Transport*, vol. 34, no. 5, pp. 600–616, 2019. https://doi.org/10.3846/transport.2019.11710

[14] J. Mitrović Simić, Ž. Stević, E. K. Zavadskas, V. Bogdanović, M. Subotić, and A. Mardani, "A novel CRITIC-fuzzy FUCOM-DEA-fuzzy MARCOS model for safety evaluation of road sections based on geometric parameters of road," *Symmetry*, vol. 12, no. 12, p. 2006, 2020. https://doi.org/10.3390/sym12122006

[15] A. A. Yürüyen, A. Ulutaş, A. Demirhan, and E. Ozsalman, "Environmental considerations in the selection of transport vehicles: A fuzzy FUCOM method," *Manag. Environ. Qual. Int. J.*, vol. 36, no. 6, pp. 1650–1669, 2025. https://doi.org/10.1108/MEQ-10-2024-0437

[16] V. Pajić, M. Andrejić, and M. Kilibarda, "Procurement optimization by selecting efficient suppliers using DEA-FUCOM-CoCoSo approach and solving order allocation problem," *Facta Univ. Ser. Mech. Eng.*, vol. 23, no. 1, pp. 109–125, 2025. https://doi.org/10.22190/FUME220210031P

[17] M. T. Le and T. D. T. Pham, "Analysis and evaluation of the impact of quantitative and qualitative factors on Vietnam's logistics efficiency using the DEA-MCDM integrated method," *Sustainability*, vol. 18, no. 3, p. 1594, 2026. https://doi.org/10.3390/su18031594

[18] J. Wang, J. Wen, V. Pajić, and M. Andrejić, "Optimizing cross-dock terminal location selection: A multi-step approach based on CI-DEA–IDOCRIW–MABAC for enhanced supply chain efficiency—A case study," *Mathematics*, vol. 12, no. 5, p. 736, 2024. https://doi.org/10.3390/math12050736

[19] N. Petrović, V. Jovanović, D. Marinković, B. Nikolić, and S. Marković, "Logistics companies' efficiency analysis and ranking by the DEA-fuzzy AHP approach," *Appl. Sci.*, vol. 15, no. 17, p. 9549, 2025. https://doi.org/10.3390/app15179549

[20] C. C. Yang, C. N. Wang, and T. T. Ngo, "Evaluating the relative efficiency of third party logistics companies using data envelopment analysis: A case study in road transportation and warehousing," *Res. Transp. Bus. Manag.*, vol. 56, p. 101181, 2024. https://doi.org/10.1016/j.rtbm.2024.101181

[21] A. Charnes, W. W. Cooper, and E. Rhodes, "Measuring the efficiency of decision making units," *Eur. J. Oper. Res.*, vol. 2, no. 6, pp. 429–444, 1978. https://doi.org/10.1016/0377-2217(78)90138-8

[22] R. D. Banker, A. Charnes, and W. W. Cooper, "Some models for estimating technical and scale inefficiencies in data envelopment analysis," *Manag. Sci.*, vol. 30, no. 9, pp. 1078–1092, 1984. https://doi.org/10.1287/mnsc.30.9.1078

[23] R. G. Thompson, F. D. Singleton Jr., R. M. Thrall, and B. A. Smith, "Comparative site evaluations for locating a high-energy physics lab in Texas," *Interfaces*, vol. 16, no. 6, pp. 35–49, 1986. https://doi.org/10.1287/inte.16.6.35